Family Outing Planner & Expense Ledger

System Architecture Document · Version 1.0 · Target Audience: Software Engineering Practice

1. Executive Summary & Problem Context

Extended family groups frequently experience friction organizing outings due to calendar coordination hurdles, consensus gridlock in group chats, and opaque expense tracking following celebrations. This system design document outlines a cross-platform mobile application built to serve as a single source of truth for family scheduling, idea voting, and financial settlement.

> **Core Architecture Strategy: Monolith First**
> To optimize for rapid MVP delivery for an initial family size (~25 users) while preserving enterprise scalability for thousands of users, the application uses a **Modular Monolith** pattern backed by Java Spring Boot and PostgreSQL.

MVP Strategy (25 Users)

- Single-instance Spring Boot container.
- Managed Neon serverless PostgreSQL.
- Zero external AI/OCR dependencies to maintain predictability.
- Minimal operational overhead & near-zero cloud cost.

Scale-Up Path (100k+ Users)

- Decouple modules into isolated microservices via Kubernetes (EKS).
- Introduce Redis for active voting consensus caching.
- Database read replicas for high-frequency calendar queries.
- Asynchronous event bus integration.

2. High-Level Technical Stack & Tooling

Layer	Technology / Package	Design Rationale
Mobile Frontend	Flutter (Dart), Hive, Dio, stomp_dart_client	Cross-platform UI with local offline-first caching (Hive), robust HTTP interceptors (Dio), and real-time STOMP WebSockets.
Backend API	Java 21, Spring Boot 3.x, Spring WebSocket	Enterprise modular monolith with virtual threads, ACID transactions, and native STOMP broker support.
Database & Hosting	Neon (Serverless PostgreSQL) & Vercel	Instant branching Postgres with strict ACID compliance, deployed via containerized Docker on Vercel.

3. Database Schema & Tenant Binding

The relational database schema is strictly partitioned by business domains. To ensure bulletproof multi-tenant isolation, every user maintains a **strict 1-to-1 relationship with a single Family** via an embedded family_id foreign key, injected into JWT claims for zero-leak queries.

Table Name	Key Attributes	Domain Purpose
Users	id, email, password_hash, family_id_fk (1-to-1)	Manages authentication and binds users exclusively to their family group.
Outings	id, title, status, budget, final_date	Tracks event planning states (draft, open, finalized, completed).
Votes	id, user_id_fk, idea_id_fk, rank	Stores ranked-choice voting entries for collaborative outing selection.
Expenses & Transactions	id, outing_id_fk, amount, payer_id_fk	Maintains ACID compliance for ledger entries and debt splits.

4. Modular Monolith Internal Architecture

Within the single Spring Boot process, code is organized strictly into self-contained domain modules communicating via clean internal interfaces and asynchronous Spring Events.

Key Architectural Refinements

- **Offline-First (Option B):** Optimistic UI updates with local SQLite/Hive caching for snappy mobile performance during brief connectivity drops.
- **Event-Based Internal Notifications:** Spring Events write directly to an internal notifications table, queried via REST/WebSockets without third-party push services.
- **Debt Simplification Ledger:** Deterministic Min-Cash-Flow greedy algorithm to minimize settlement transfers.

5. Core Feature Trade-Off Analysis

Every feature added to the system carries architectural costs and benefits weighed against MVP velocity.

Feature	Architectural Benefit	Trade-Offs & Costs
Multi-Calendar Overlap	Eliminates group chat scheduling gridlock by aggregating availability.	Requires internal state management or complex OAuth provider integrations.
Debt Simplification Ledger	Reduces complex multi-person receipts into minimal settlement transactions.	Demands strict ACID transaction integrity to prevent currency discrepancies.
Ranked-Choice Voting	Provides democratic consensus for outing selection.	Adds state machine complexity for tallying and finalizing votes.

6. UI Screen Workflow Architecture

The mobile user experience guides family members through three core interconnected workflows designed to solve recurring planning pain points:

Primary User Journeys

- **Planning & Consensus:** Dashboard → Create Outing → Set Availability → Pitch Ideas → Ranked-Choice Voting → Finalized Event.
- **Expense Management:** Outing Details → Add Expense → Outing Ledger Overview → Debt Settlement Settle-Up Screen.